tsdown 

One Tool to Bundle Them All



2025-10-09

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An elegant library bundler



2025-10-09

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Maintainer of Unplugin / Quansync

Sponsored by VoidZero



sxzz sxzz.dev sanxiaozhizi



What is Library Bundler?

🔍 Essence



All kinds of source code

TS / Vue SFC / JSX / CSS...





Library outputs ready to use

ESM / CJS / .d.ts / Minified / Tree Shaking

🔎 Essence



All kinds of source code

TS / Vue SFC / JSX / CSS...





Library outputs ready to use

ESM / CJS / .d.ts / Minified / Tree Shaking

🚀 Your code can be used directly by other projects



Why Another Library Bundler?



Why Another Library Bundler?

🤦 Rollup, esbuild, tsup, unbuild...



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🤦 Rollup, esbuild, tsup, unbuild...

Why do we need tsdown?

🤕 Pain Points

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Complicated Configuration

Rollup / esbuild configs are complex and error-prone

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Type Declarations

tsc is hard to integrate, need custom build scripts

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Multi-format Output

ESM / CJS / IIFE configs are inconsistent

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Multi-format Output

ESM / CJS / IIFE configs are inconsistent



Performance Bottlenecks

Slow builds for large libraries

🎯 Purpose

Born for modern library development

- Minimal configuration, ready to use out of the box
- Automatic type declarations and dependency analysis
- Consistent multi-format output
- Extreme performance, powered by 🦀 Rust
- Embraces modern frontend development paradigms

💁 What Can tsdown Do?

💁‍♂️ What Can tsdown Do?



Node.js / CommonJS

```
require('fs')
```

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Vue Components

`App.vue`

💁 What Can tsdown Do?



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Minification

Minification / Tree-shaking

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Bundling

bundle / bundleless

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Powerful Plugins

unplugin / Rollup / Vite

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Powerful Plugins

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😎 And much more—too many to list!

🏄 Progressive Start

Create a new project

```
pnpm create tsdown@latest
```

Manual install

```
pnpm i -D tsdown
```

Bundle with one command

```
tsdown
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- Supports npm / pnpm / Yarn / Bun
- Node.js 20.19+; Deno / Bun experimental support
- CLI and config file can be mixed, e.g. `tsdown --no-dts`
- All common options supported, like `--format`, `--watch`

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- Performance first
  - Supports `isolatedDeclarations`, powered by Oxc
  - Supports TypeScript Go (tsgo.sxzz.dev)
  - Supports multi-project, multi-threaded generation

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  - Supports TypeScript Go (tsgo.sxzz.dev)
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- Supports Vue components

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Multi-format Output & Platform Support

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ESM



CJS



IIFE



UMD

Multi-format Output & Platform Support



ESM



CJS



IIFE



UMD

- Output multiple formats with one command

```
tsdown --format esm,cjs
```

Multi-format Output & Platform Support



ESM



CJS



IIFE



UMD

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- Supports `node` / `browser` / `neutral` platforms

Multi-format Output & Platform Support



ESM



CJS



IIFE



UMD

- Output multiple formats with one command

```
tsdown --format esm,cjs
```

- Supports `node` / `browser` / `neutral` platforms
- Read `package.json#engines` for syntax lowering (`a?.b` / `a &&= b`)

Auto-generate `package.json#exports`

```ts
export default defineConfig({
  exports: true,
})
```

or

```
tsdown --exports
```

Auto-generate `package.json#exports`

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export default defineConfig({
  exports: true,
})
```

or `tsdown --exports`

- Automatically generates `main/module/types/exports` fields in your `package.json`
- Points to source code during development, switches to build output for publishing
- Custom exports rules supported

Vue Support

```ts
// tsdown.config.ts
import Vue from 'unplugin-vue/rolldown'

export default defineConfig({
  plugins: [Vue()],
  dts: { vue: true },
  platform: 'neutral',
})
```

Plugin Ecosystem & Extensibility

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Rolldown Plugins



Unplugin Plugins



Rollup Plugins

Plugin Ecosystem & Extensibility



Rolldown Plugins



Unplugin Plugins



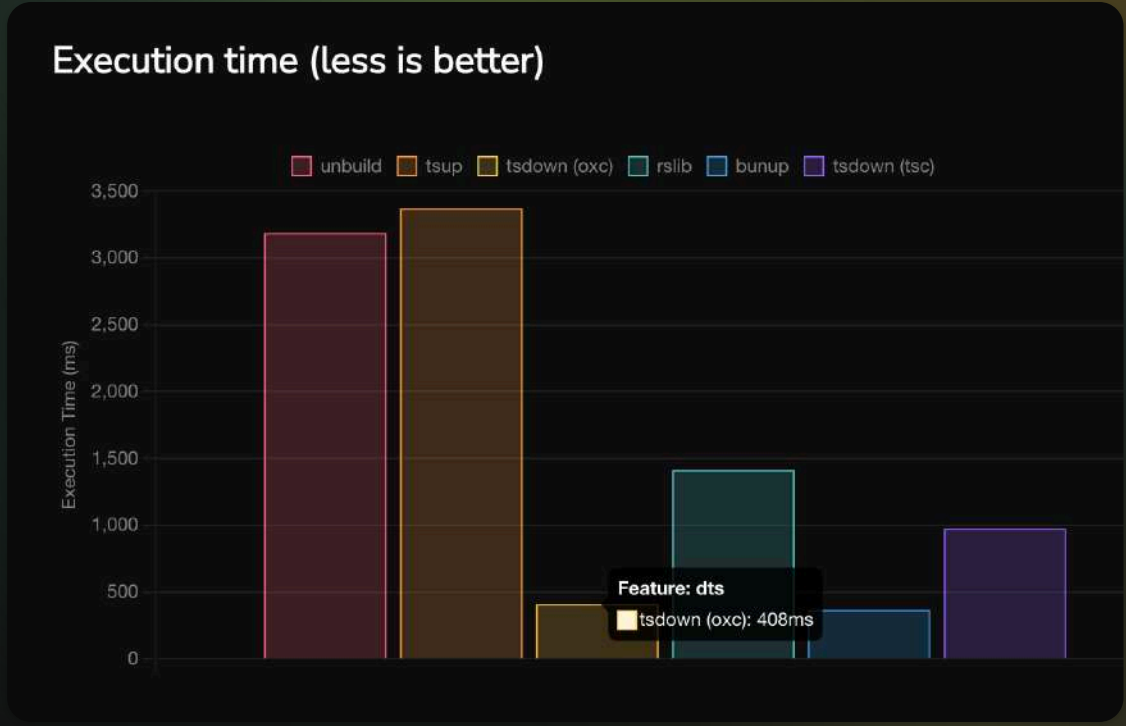
Rollup Plugins



Vite Plugins

Performance

Fast! Really fast!



Gugustinette/bundler-benchmark

Migrating from tsup is Super Easy

```
tsdown migrate
```



Auto-migrate with one command

Highly compatible config

Feature Overview



Tree-shaking



Minification



Source Map



Watch Mode



Unbundle Mode



Shims



Build Report



Silent Mode



Plugin System



Auto Dependency Analysis



Auto Type Declarations



Multi-entry / Multi-output



Rolldown Extreme Performance



Vite/Vitest Config Reuse



One-click tsup Migration



Monorepo Support

tsdown Status

As of 2025-10-07

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873K NPM downloads / month

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52 Contributors

tsdown Status

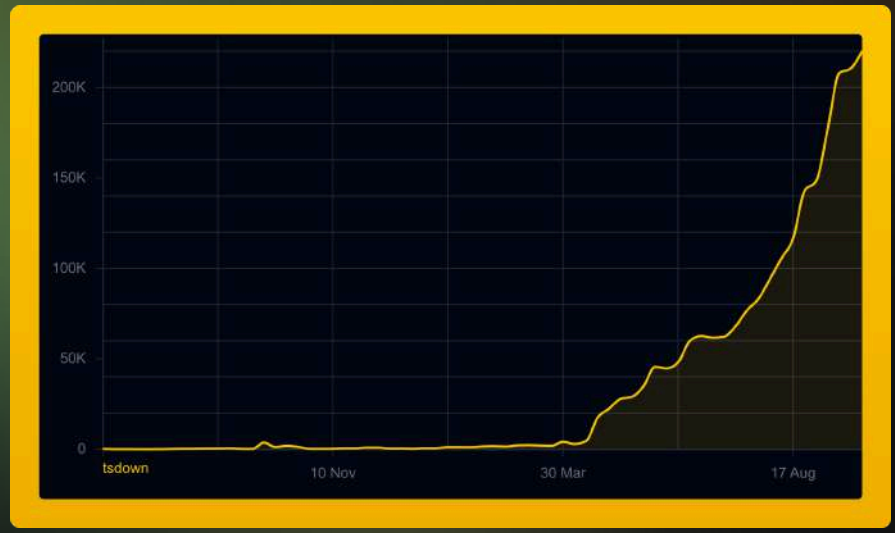
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Next Steps

Vite Library Mode Integration

- Aligning options with Vite's library mode (e.g., `logLevel`, `configLoader`)
- Full support for Vite plugins

Shoulders of Giants

tsup

Rolldown

esbuild

Vite

`rollup-plugin-dts`

unbuild

Oxc

Rollup

unplugin

...



rolldown/tsdown



rolldown/tsdown



tsdown.dev

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